

Large Language Models

Final Exam – Session 2025-26

Master MIND – Sorbonne Université

Duration: 2 hours

Part I: Course Questions

Transformer Architecture

Question 1 – Self-Attention Mechanism

[9 pts]

For two tokens i and j , self-attention computes:

$$p(i|j) \propto \exp((Q_h x_i) \cdot (K_h x_j))$$

- 1.1) Multi-head attention uses h parallel attention heads with smaller dimensions ($d_k = d_{\text{model}}/h$). Beyond computational efficiency, what is the benefit of multiple heads compared to a single attention with full dimension?
- 1.2) Consider the input sequence “1 + 2 = [MASK]” where we want to predict the token “3” at position [MASK]. What is the minimum number of self-attention layers required to solve this task? Justify your answer by explaining what information needs to flow between positions.
- 1.3) In a decoder (autoregressive model like GPT), a causal mask prevents position i from attending to positions $j > i$. Why is this necessary during training, and how does it enable parallel computation?

Training & Efficiency

Question 2 – Pre-training Objectives

[6 pts]

Masked Language Modeling (MLM) corrupts 15% of tokens and predicts them:

$$\mathcal{L}_{\text{MLM}} = - \sum_{i \in \text{masked}} \log p_{\theta}(t_i | \tilde{t}_{1:L})$$

Causal Language Modeling (CLM) predicts the next token:

$$\mathcal{L}_{\text{CLM}} = - \sum_{i=1}^L \log p_{\theta}(t_i | t_{1:i-1})$$

- 2.1) MLM uses bidirectional context while CLM uses only left context. Explain why this makes MLM-trained models (like BERT) unsuitable for text generation without modification.

- 2.2) We want to train an autoregressive (causal) language model but using the MLM objective instead of CLM. Describe how you would modify the training setup (masking strategy, attention mask, or both) to achieve this.

Question 3 – LoRA (Low-Rank Adaptation)

[5 pts]

LoRA adds trainable low-rank matrices to frozen weights:

$$h = W_0x + \Delta Wx = W_0x + BAx$$

where $W_0 \in \mathbb{R}^{d \times d}$ is frozen, $B \in \mathbb{R}^{d \times r}$, $A \in \mathbb{R}^{r \times d}$, and $r \ll d$ (typically $r = 4$ to 16).

- 3.1) B is initialized to zeros and A with random values. Explain why.
 3.2) LoRA with $r = 8$ achieves similar accuracy to full fine-tuning on many tasks. Does this mean the task adaptation only requires an 8-dimensional subspace? Explain.

Question 4 – Flash Attention

[3 pts]

Standard attention computes $\text{softmax}(QK^T)V$ by:

1. Computing $S = QK^T$ (size $N \times N$ for sequence length N)
2. Computing $P = \text{softmax}(S)$
3. Computing output $O = PV$

Each intermediate matrix S and P must be stored in GPU high-bandwidth memory (HBM).

Flash Attention processes attention in blocks using *incremental softmax*.

- 4.1) Explain how Flash Attention can compute the exact same output as standard attention without ever materializing the full $N \times N$ attention matrix in memory.

Generation & In-Context Learning

Question 5 – Decoding Strategies

[2 pts]

- **Greedy:** always select $\arg \max p(t_i | t_{<i})$
- **Temperature sampling:** sample from $p(t_i | t_{<i})^{1/T}$ (renormalized)
- **Top-p (nucleus) sampling:** sample from the smallest set of tokens whose cumulative probability exceeds p

Top-p sampling with $p = 0.9$ adapts the number of candidate tokens based on the distribution's shape.

- 5.1) When would nucleus sampling select very few tokens?
 5.2) When would it select many tokens?

Question 6 – In-Context Learning (ICL)

[6 pts]

ICL allows models to perform tasks by including examples in the prompt, without updating parameters.

- 6.1) Experiments show that few-shot prompting with *random labels* (e.g., labeling positive reviews as “negative”) still improves over zero-shot performance. What does this suggest about how ICL works?
- 6.2) Chain-of-Thought (CoT) prompting includes reasoning steps in examples: “Q: ... Let's think step by step: [reasoning]. Answer: X”. This helps on arithmetic but not on sentiment analysis. Explain why.

Neural Information Retrieval & RAG

Question 7 – Dense vs. Sparse Neural Retrieval

[8 pts]

Bi-encoders encode query and document independently:

$$\text{score}(q, d) = \text{enc}(q)^T \text{enc}(d)$$

SPLADE produces sparse vectors using MLM logits:

$$w_j^{(d)} = \max_i \log(1 + \text{ReLU}(\text{MLM}_{\text{logits}}[i, j]))$$

where j indexes vocabulary terms and i indexes document positions.

- 7.1) Bi-encoders enable pre-computing document embeddings for fast retrieval. Why can't cross-encoders (which process query and document jointly) benefit from this?
- 7.2) SPLADE produces vectors with >99% zeros. Why is high sparsity desirable for retrieval, and what trade-off does it introduce?
- 7.3) SPLADE is trained with a FLOPS regularization loss that penalizes dense activations. What would happen if this regularization were removed?

Question 8 – Retrieval-Augmented Generation (RAG)

[3 pts]

RAG retrieves documents and includes them in the prompt before generation.

- 8.1) A RAG system sometimes produces worse answers when retrieval finds partially relevant documents (compared to no retrieval). Explain why this can happen, and give an example scenario where retrieval could degrade performance.

Alignment & Interpretability

Question 9 – Bias and Steering

[8 pts]

Activation steering modifies hidden states during inference:

$$h' = h + \alpha \cdot v_{\text{steer}}$$

where v_{steer} is a “steering vector” computed as the difference between activations on contrasting examples.

- 9.1) To reduce gender bias, how can the steering vector be computed?

- 9.2) This steering vector was computed on gender-related sentences. Would it also reduce racial bias? Justify your answer.
- 9.3) Name and briefly describe the main approaches used to align a language model with human preferences or values.

Part II: Paper Analysis

Question 10 – Paper Comprehension

[20 pts]

- 10.1) The LRE approximation $F(s, c_r) \approx \beta W_r s + b_r$ involves a first-order Taylor expansion, yet transformers are highly non-linear. Identify and explain the two specific architectural features that cause this linear approximation to lose accuracy in magnitude even when it captures the correct direction of change. —
- 10.2) Why do the authors need *both* faithfulness and causality to validate an LRE? Construct a hypothetical scenario where an LRE scores high on one metric but low on the other, and explain what this would reveal about the model's internal computation. —
- 10.3) The relation *company CEO* achieves under 6% faithfulness despite GPT-J correctly predicting CEOs for 69 companies. Does this mean the model lacks this knowledge, or that it is encoded differently? Justify your answer using evidence from the paper.
- 10.4) In the attribute lens experiment (Section 5), the model outputs a falsehood (e.g., “Oslo”) yet the attribute lens recovers the true fact (“London”) from intermediate representations. Explain the mechanism that makes this possible and why the logit lens would fail in the same setting.

Question 11 – Critical Analysis

[20 pts]

- 11.1) The causality experiment patches a single hidden state at layer ℓ_r , but earlier layers still retain information about the original subject and can leak it forward via attention. Design a stronger causal intervention that would control for this confound, and discuss its trade-offs.
- 11.2) The correlation between faithfulness and causality is $R = 0.84$ when hyperparameters are selected to maximize causality, but only $R = 0.74$ when optimizing for faithfulness (Figure 10). What does this asymmetry tell us about the relationship between the two metrics? —
- 11.3) LREs are estimated from $n = 8$ examples, yet Figure 12 suggests performance plateaus around $n = 5$. Is this surprisingly few? Discuss what this implies about the dimensionality of the relational subspace the Jacobian needs to capture.
- 11.4) The attribute lens can detect internally-known facts that the model does not output. Explain why this is the case, and propose a concrete system design that exploits this capability for **hallucination detection** at inference time, and discuss its practical limitations.